TEMPERATURE MONITORING NEURAL NETWORK USING ReLU

Project Title: Temperature Monitoring Neural Network Using ReLU

Application: Temperature Monitoring

Concept: ReLU Activation Function

Dataset: Temperature Dataset

1. Problem Statement

A neural network for temperature monitoring frequently receives negative pre-activation values. Implement a simple neuron using ReLU and test it with positive and negative temperature-related input values. Analyze the resulting activation outputs, compare the responses for positive and negative inputs, and determine how ReLU affects the representation of the input values.

2. Objective

- Implement a simple artificial neuron for temperature inputs.
- Calculate the pre-activation value using weight and bias.
- Apply the ReLU activation function.
- Test the neuron using positive and negative temperature values.
- Compare negative and positive pre-activation responses.
- Analyze how ReLU affects the representation of input values.
- Generate visualizations of the activation behavior.

3. Concept Used

The artificial neuron uses the linear transformation: $z = w \times x + b$. Where: x = temperature input; w = 0.5; b = -8. The pre-activation value is: $z = 0.5x - 8$. The ReLU activation function is: $ReLU(z) = \max(0,z)$. ReLU converts negative values to zero, passes positive values unchanged, introduces non-linearity, and produces sparse activations.

3.1 ReLU Activation Function

Property	Description
Negative pre-activation	Converted to zero.
Positive pre-activation	Passed through unchanged.
Zero pre-activation	Remains zero.
Non-linearity	Produces a piecewise linear response.
Sparsity	Produces zero activation for negative values.

4. Methodology

Input: Load temperature values from dataset/temperature_data.csv.

Neuron parameters: Set weight w = 0.5 and bias b = -8.

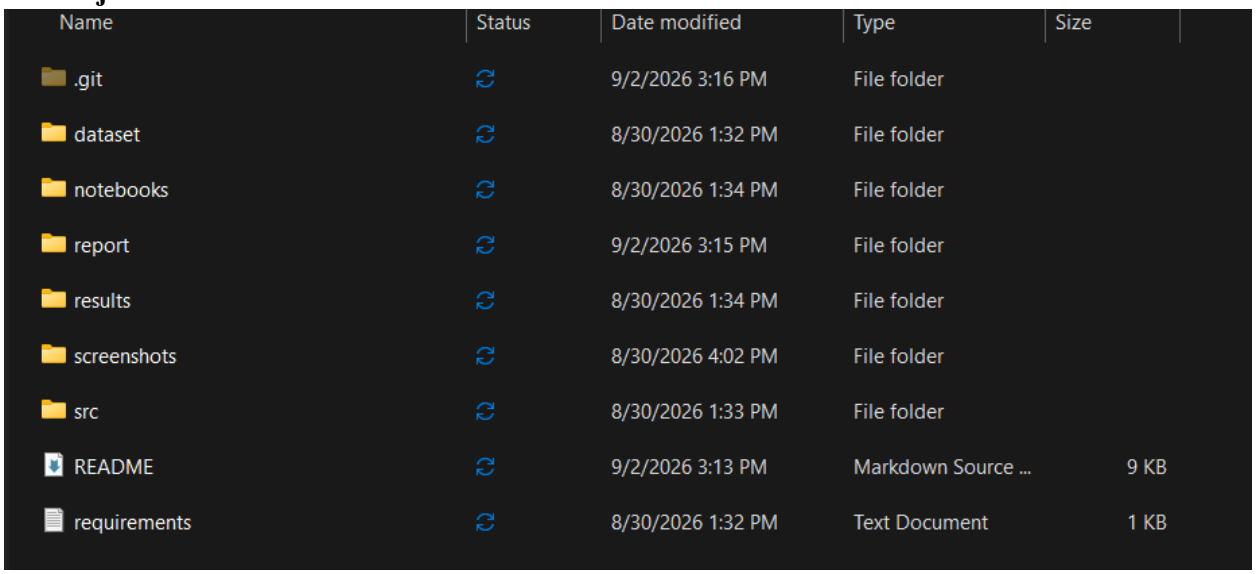
Weighted input: Calculate w * x.
Pre-activation: Calculate z = wx + b.
ReLU activation: Calculate ReLU(z) = max(0,z).
Analysis: Compare the pre-activation values with the ReLU outputs.
Sparsity: Count zero and non-zero activation outputs.
Visualization: Generate graphs showing temperature, pre-activation, and ReLU output.
Output: Save the generated visualizations and execution screenshot.

5. Implementation

5.1 Tools & Libraries

Tool / Library	Use
Python	Core programming language
NumPy	Numerical calculations and ReLU operation
Pandas	Loading and processing CSV data
Matplotlib	Visualization and plotting
Jupyter Notebook	Interactive step-by-step analysis

5.2 Project Structure



5.3 Source Code / Github Repository

URL: https://github.com/HariM917/temperature-relu-neuron
The complete source code, dataset, Jupyter notebook, generated results, README documentation, and execution screenshot are maintained in the GitHub repository.

6. Results & Output

The program implements a single temperature-monitoring neuron using a weight of 0.5 and a bias of -8. The pre-activation is calculated using z = wx + b, followed by ReLU activation.

Temp (°C)	Weighted Input	Bias	Pre-activation (z)	ReLU Output
-20	-10.0	-8	-18.0	0.0
-10	-5.0	-8	-13.0	0.0
0	0.0	-8	-8.0	0.0
5	2.5	-8	-5.5	0.0
10	5.0	-8	-3.0	0.0
20	10.0	-8	2.0	2.0
30	15.0	-8	7.0	7.0
40	20.0	-8	12.0	12.0

Output Summary

- Total inputs: 8
- Zero outputs: 5
- Non-zero outputs: 3
- Sparsity: 62.5%

Example calculations:

For -20°C: z = (0.5 × -20) - 8 = -18. ReLU(-18) = 0.

For 40°C: z = (0.5 × 40) - 8 = 12. ReLU(12) = 12.

Threshold: 0.5x - 8 = 0, therefore x = 16°C.

FIGURES

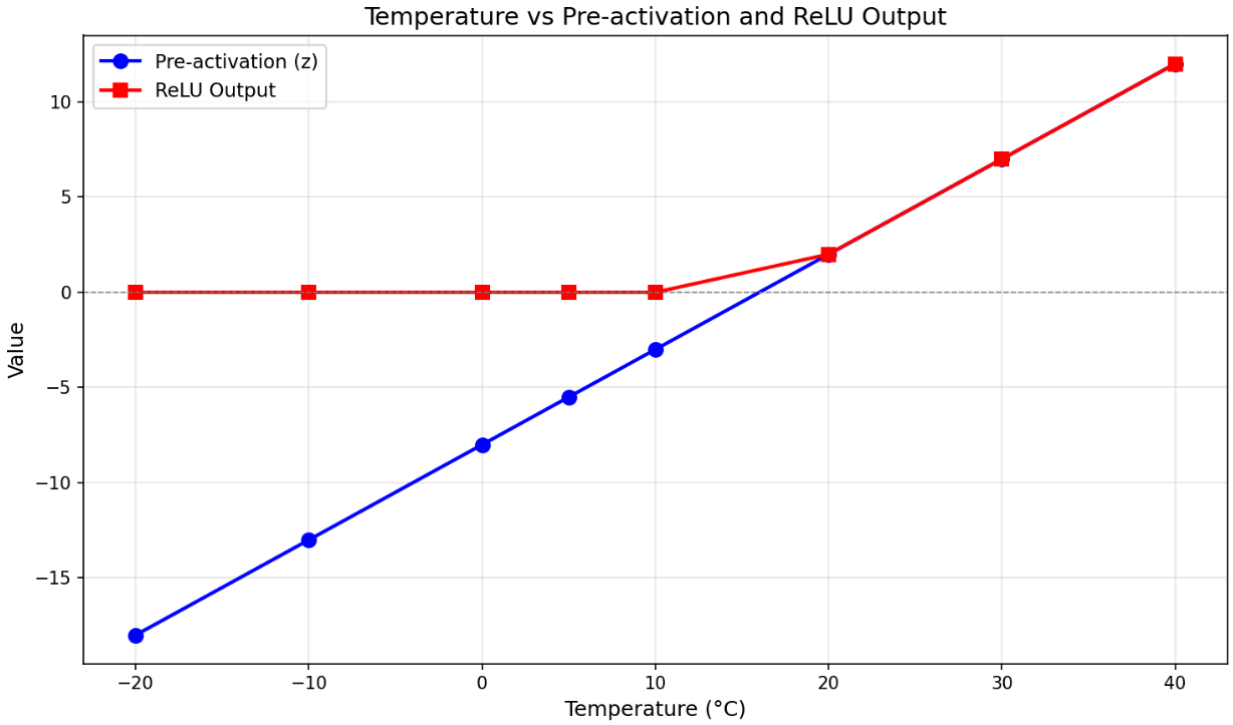


Figure 1: Temperature versus pre-activation and ReLU output.

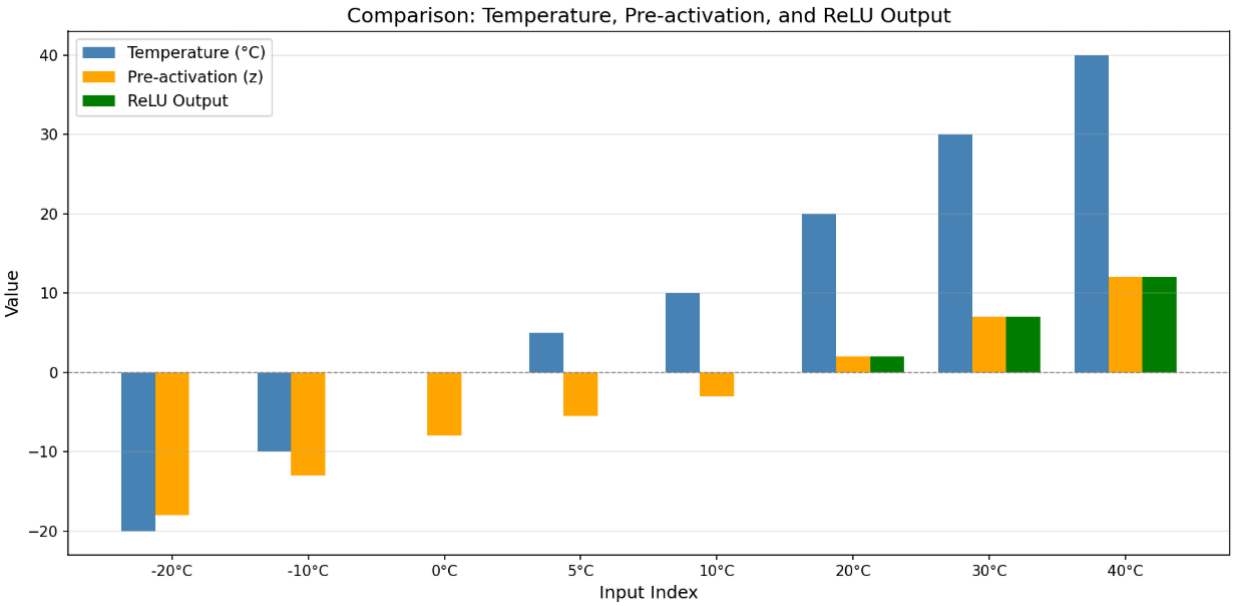


Figure 2: Comparison of temperature, pre-activation, and ReLU output.

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PS C:\Users\HARI> cd "C:\Users\HARI\OneDrive\Desktop\DLE IA\temperature-relu-neuron"
PS C:\Users\HARI\OneDrive\Desktop\DLE IA\temperature-relu-neuron> python src\relu_temperature_neuron.py
Loaded 8 temperature values: [-20 -10   0   5  10  20  30  40]

======================================================================
TEMPERATURE NEURON ACTIVATION RESULTS
======================================================================

Neuron Parameters:
  Weight = 0.5
  Bias   = -8

Formula:
  z = 0.5 * temperature + (-8)

Activation:
  ReLU(z) = max(0, z)

----------------------------------------------------------------------
  Temperature (°C) |  Weighted Input |   Bias |  Pre-activation |  ReLU Output
----------------------------------------------------------------------
            -20.0 |           -10.0 |   -8.0 |           -18.0 |          0.0
            -10.0 |            -5.0 |   -8.0 |           -13.0 |          0.0
              0.0 |             0.0 |   -8.0 |            -8.0 |          0.0
              5.0 |             2.5 |   -8.0 |            -5.5 |          0.0
             10.0 |             5.0 |   -8.0 |            -3.0 |          0.0
             20.0 |            10.0 |   -8.0 |             2.0 |          2.0
             30.0 |            15.0 |   -8.0 |             7.0 |          7.0
             40.0 |            20.0 |   -8.0 |            12.0 |         12.0
----------------------------------------------------------------------

Summary:
  Total inputs:      8
  Zero outputs:      5 (neuron did NOT fire)
  Non-zero outputs:  3 (neuron fired)
  Sparsity:          5/8 = 62.5%
======================================================================

Plot saved: C:\Users\HARI\OneDrive\Desktop\DLE IA\temperature-relu-neuron\results\relu_activation_plot.png
Plot saved: C:\Users\HARI\OneDrive\Desktop\DLE IA\temperature-relu-neuron\results\temperature_activation_comparison.png

Done! Check the 'results/' folder for saved plots.
PS C:\Users\HARI\OneDrive\Desktop\DLE IA\temperature-relu-neuron>
```

Figure 3: Program execution showing neuron activation results and summary analysis.

7. Analysis

- The experiment demonstrates that ReLU suppresses negative pre-activation values by converting them to zero.

- For temperatures from -20°C to 10°C, the calculated pre-activation values are negative and therefore produce zero ReLU outputs.

- For temperatures 20°C, 30°C, and 40°C, the pre-activation values are positive and ReLU passes them unchanged.

- The threshold for the selected weight and bias is 16°C.

- Five out of eight inputs produce zero activation.

- The resulting sparsity is 62.5%.

- The experiment demonstrates that ReLU changes the representation of the pre-activation values by suppressing negative responses while preserving positive responses.

8. Conclusion

This mini project demonstrates the use of the ReLU activation function in a simple temperature-monitoring neuron. The neuron calculates the pre-activation using $z = wx + b$ and applies ReLU to the result. Negative pre-activation values are converted to zero, while positive values are passed unchanged.

For the selected dataset, five of eight inputs produce zero outputs, resulting in 62.5% sparsity. The project demonstrates the thresholding, non-linearity, and sparse activation behavior of ReLU.

9. References

1. Nair, V., & Hinton, G. E. (2010). Rectified Linear Units Improve Restricted Boltzmann Machines. ICML-10.
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
3. Glorot, X., Bordes, A., & Bengio, Y. (2011). Deep Sparse Rectifier Neural Networks. AISTATS.
4. NumPy Documentation - https://numpy.org/doc/
5. Pandas Documentation - https://pandas.pydata.org/docs/
6. Matplotlib Documentation - https://matplotlib.org/stable/contents.html
7. Project GitHub Repository - https://github.com/HariM917/temperature-relu-neuron